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Trajectory Matters: Impact of Jamming Attacks Over the Drone Path Planning on the Internet of Drones

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ABSTRACT

The Internet of Drones (IoD) is a network paradigm where drones will fly over well-defined airways. The Jamming Attack (JA) poses a severe risk to IoD, affecting the drone's trajectory. Although JA has been investigated in UAV-based networks, the current solutions consider the airspace free to fly, whereas IoD allows drones to fly over constrained airspace, named airways. This study investigates the impact of JA over IoD, mainly regarding the drone path planning and, therefore, the drone trajectory. Also, we conduct a thorough discussion regarding the current challenges. To overcome these challenges, we propose the IoD-JAPM, an airway-aware protection mechanism against JA on the IoD, ranging from analyzing the airway's availability to the potential reformulation of the drone path planning. We perform an experimental evaluation through simulations to compare IoD-JAPM with a baseline solution and an existing approach that considers the airspace free to fly, varying the IoD topology and the number of jammers in the environment. We analyze four different metrics to measure at what level IoD-JAPM can mitigate a JA. Through the results, we observe a relation between the impact of JA on the drone path planning and the robustness of the topology, where the more restricted the topology, the more impact JA causes. Also, IoD-JAPM mitigates the effects of JA on path planning, causing few reformulations or even flight cancellations. Specifically, IoD-JAPM prevents all drones from being attacked when the environment has one jammer and mitigates the attack at 30% for two jammers. Furthermore, IoD-JAPM overcomes the baseline study regarding the increasing drone flight distance, and power consumption. Since the drone propellers affect most of the drone battery, the more the flight distance the more the power consumption. Also, the results reinforce that "free-to-fly" approaches cannot be applied in airway-oriented environments with strong jamming signals. In a nutshell, IoD-JAPM represents a robust protection mechanism against JA considering the aerial space defined through airways, advancing the state of the art of IoD-based protection mechanisms against JA.

1. Introduction

Drones have been awakening the interest of both industry and academia. The commercial use of drones has been changing considerably, in which their autonomous use (without the control of a human pilot) is increasingly common, providing a better quality of service (QoS) [1,2] and leveraging an enhanced scenario of Intelligent Transportation Systems (ITS) in urban environments [3,4].

In this scenario, drones will dispute the airspace and the wireless communication channel. Hence, it will be mandatory to have a system to manage and provide a fair and reliable environment, named the Internet of Drones (IoD). Currently, there are some efforts to address a real IoD scenario [5]. IoD can be modeled as a layered network architecture for coordinating the access of drones to the airspace, providing navigation services between Points of Interest (PoIs) [5].

One of the most significant characteristics of this paradigm is the concept of well-established airways. They are similar to grounded roads, having several traffic policies, for instance, airspace boundaries and speed limits. Indeed, airways will play a fundamental role in the near future, when it is expected that a massive number of drones will fly over people. Hence, drones must follow a well-defined path planning [5]. Furthermore, drones have considerable Size, Weight, and Power (SWaP) limitations, which substantially affect their flight autonomy.

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IoD is a mobile network paradigm in its early stages, demanding infrastructure deployment and regulatory guidelines. Regarding the infrastructure, a *vertiport* is an "airport" for Vertical Take-off and Landing (VTOL) vehicles, such as drones [6]. In the IoD context, *vertiports* can be considered a PoI shared by drones (e.g., a building) to assist them in some tasks, such as battery recharge and delivery pick-up.

Different attacks threaten IoD security. Since it represents a robust assisting environment for other networks – for instance, Cellular and Vehicular Adhoc Networks (VANETs) – drones are the primary targets of malicious adversaries, suffering different attacks [7–9]. Some of them can hamper the availability of a targeted node, hindering it from performing its tasks. For instance, Jamming Attack (JA) aims to make a node unavailable in the network. The attack comes from single or multiple malicious entities that congest the communication channel, precluding the targeted node communicates with other legitimate nodes [10].

1.1. Problem statement

JA represents a severe risk in IoD because availability is a paramount requirement in this network. There are some effective countermeasures against JA in terrestrial mobile networks, such as Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS) [11]. However, they are ineffective in IoD because they cannot appropriately handle LoS-induced security issues [12].

Once a drone has its availability affected, it can present an unusual behavior, such as landing at inappropriate places, leading to other attacks, such as hijacking [13]. In some situations, an airway can have all its flyable airspace affected by a JA, making the drones that fly over there incommunicable [14,15]. These issues can also affect the trajectory of drones, leading to a revised path planning.

The relation between JA and the drone trajectory has been widely discussed in UAV-based networks [14–22], but those studies consider the airspace free to fly.

In contrast, one of the most prominent characteristics of IoD is the presence of airways, allowing drones to fly over constrained airspace. Recent studies consider IoD organizing airspace through airways, demonstrating their importance for adequately managing the flight of drones [9,23–26]. Since the airways limit this flyable airspace, current solutions cannot be applied directly to IoD.

To the best of our knowledge, our previous study was the first to investigate the impact of JA over constrained airspace, introducing a mechanism to avoid JA in the IoD [27]. In a nutshell, the proposed mechanism identifies and isolates a region affected by JA, reformulating the path planning of drones. Although it provided adequate protection to the drones against JA, it also revealed that constrained airway topologies tend to hamper the provided protection. Moreover, whenever the number of jammers increases, there is an increment in the flight distance. Also, the mechanism cannot handle situations when the drone destination belongs to a region suffering an active JA. All these issues represent challenges to investigate.

Bearing these challenges in mind, this study deals with how JA impacts the drone trajectory and path planning. Notice that IoD allows drones to fly exclusively over constrained airspace, defined through the airways. Furthermore, this study embraces the challenges highlighted in our previous study [27] (Sections 2.2 and 3.2) and how to overcome them (Section 4).

1.2. Goals and contributions

This study investigates the impact of JA on drone path planning and, therefore, its trajectory. We conduct a thorough discussion regarding the current challenges in this field. To overcome these challenges, we propose the IoD-JAPM, an airway-aware protection strategy against JA on the IoD. The mechanism ranges from analyzing the airway's availability to reformulating the drone path planning.

IoD-JAPM enhances our previous protection solution [27] by considering the following aspects: the LoS communication between drones and the IoD base stations, named Zone Service Providers (ZSPs) [5]; and the presence of *vertiports* over the IoD environment.

We list our main contributions as follows:

► We model a method to isolate a region affected by a JA in the IoD, considering the airways topology;

► We propose a strategy to avoid a jamming signal in an affected region without violating the flight restrictions imposed by the airways. This strategy considers the LoS communication and the available airspace between two parallel airways with different altitudes;

► We propose a strategy to mitigate the impact of the reformulated drone path planning when its final destination belongs to a region affected by the JA. This strategy considers the presence of *vertiports* in the environment.

All these contributions overcome the challenges presented in our previous study [27]. Furthermore, we discuss the impact of JA in the IoD environment, mainly regarding the drone trajectory. To analyze the robustness of IoD-JAPM, we carried out a comparative evaluation with our previous study [27] and with an existing approach that considers the airspace free to fly, primarily [18]. Therefore, we advance the state of the art of IoD-based protection mechanisms, handling its specific requirements.

This work is organized as follows: Section 2 presents a background of IoD and JA concepts. Section 3 discusses the related work. The proposed protection mechanism, IoD-JAPM, is described in Section 4. After, Section 5 details the experimental evaluation, including the evaluated scenario, simulation parameters, and metrics. Section 6 provides the obtained results as well as a thorough analysis. Finally, Section 7 offers our final remarks.

2. Background

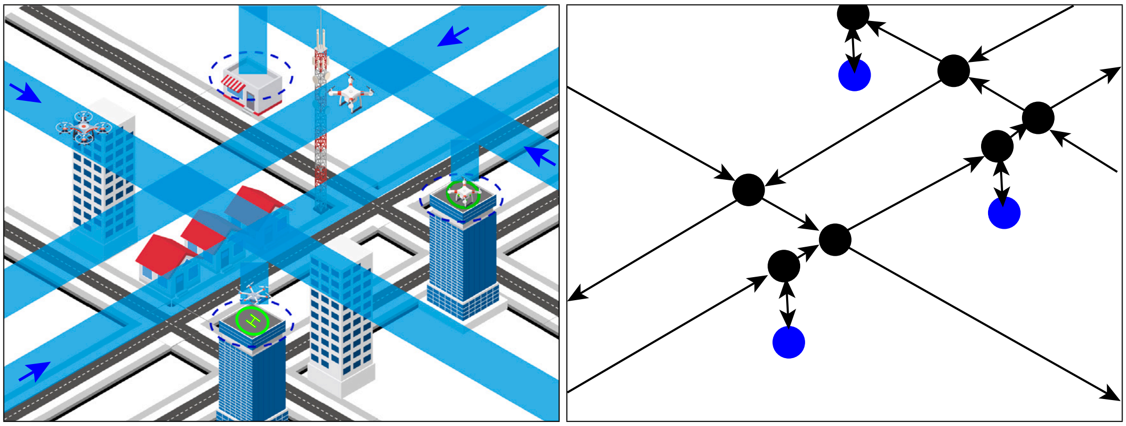
This section presents the main concepts regarding the IoD as an ITS scenario, and the concepts about JA.

2.1. Internet of drones

IoD is the concept of a robust network to manage and control airspace, providing navigation services to drones. IoD can support and integrate with grounded networks, expanding the IoT coverage. Given this perspective, IoD is proposed as a layered network control architecture, where the drones fly over the airways having well-defined path plannings [5]. The ZSP manages and provides navigation information. Since drones move fast through the airspace, this location update time interval must be small. Therefore, the IoD cloud system can properly manage the access and navigation of drones in such a way as to avoid collisions and traffic jams.

This environment can be modeled as a graph $G = (V, E)$. V is a set of 3-Dimensional waypoints the drones must reach based on their path planning. E is the set of airway segments, delimited by two waypoints $w_1, w_2 \in V$, such that $\langle w_1, w_2 \rangle \in E$. Each edge $e \in E$ has some information, for instance, the maximum speed allowed, the maximum number of concurrent drones inside it, and aerial boundaries, commonly represented by a radius r. Hence, a drone path planning is a subgraph $G' = (V', E') \subset G$. During a flight, a drone must send its current location periodically to the nearest ZSP.

Fig. 1 illustrates the IoD environment as an envisioned ITS scenario. Fig. 1(a) represents an urban scenario where the blue segments are the flyable airways. As we can note, they are parallel to the grounded roads. Furthermore, these environment has PoIs, such as *vertiports* and delivery pick-up regions (represented by the dashed blue circles). Drones use dedicated airways to access these PoIs. Fig. 1(b) depicts how this topology can be modeled as a graph. Given two intersection nodes, u and v (represented by the black circles), a directional edge indicates how a drone can fly between the nodes. Blue circles indicate the nodes related to PoIs.



(a) IoD in an urban scenario. Blue segments represent the airways parallel to the grounded roads. The Blue dashed circles indicate PoIs, such as *vertiports*

(b) IoD airways' topology represented as a graph. Black circles represent intersection nodes. Blue circles represent PoI nodes. Black arrows represent the edges, indicating a flyable airway between two nodes.

Fig. 1. IoD environment as an ITS.

2.2. Jamming attack

A Jamming Attack (JA) is a type of Distributed Denial of Service (DDoS) affecting the nodes' availability in the environment. JA occurs when adversaries interfere in the communication between a set of network nodes by flooding the network communication channels, occupying a part of (or the whole) bandwidth. Hence, the nodes cannot use the network because the communication channel is always occupied [10].

In JA, multiple stationary adversaries can be distributed over the environment. They will perform the attack considering the best configuration for them. A given adversary can perform short-term and long-term attacks. An attacker also cannot perform a JA. Thus, given an attacker α and a time interval Δ_t whose initial time is t_0 and the last time is t_n, these attacks can be characterized as the following [10]:

- *Absence of JA*: $\sum_{k=0}^{n} J_k = 0$;
- *Long-term JA*: $\forall k \in \langle 0, n\rangle : J_k \neq 0$, meaning the attack remains over all Δ_t;
- *Short-term JA*: given a time threshold t_i, $\sum_{k=0}^{i} J_k = 0$ and $\sum_{k=i+1}^{n} J_k \neq 0$, or vice-versa. In other words, the attack is performed in a short period of time. The subsequent occurrence of this attack indicates that the adversary is performing an intermittent JA.

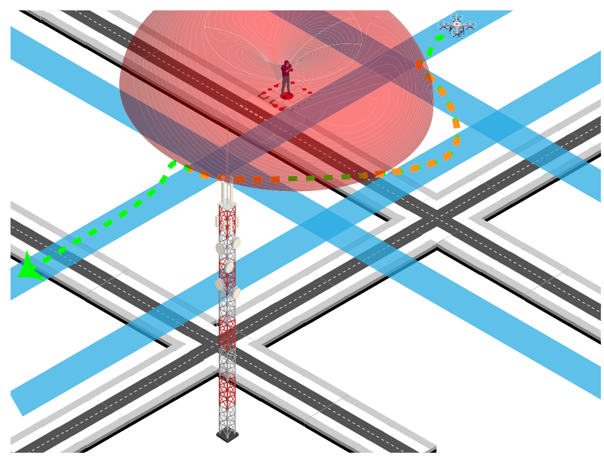


Fig. 2. An example of how the current solutions to avoid a JA is not applicable in an environment with well-defined airways. Dashed orange lines indicate an invasion of the drone in unauthorized airways.

3. Related work

This section discusses the recent studies that address the JA's impact on the drone trajectory. Since this is the first study to analyze how JA impacts the path planning of drones following well-defined airways, the studies mentioned in this section are partially related to our solution.

3.1. Mechanisms to avoid JA on UAV-based networks

In recent years, JA has been investigated in the context of the UAV-based network as a threat and security mechanism [28]. Specifically, some studies focused on analyzing how JA interferes with the drone trajectory. The majority of these proposals consider a grounded and stationary jammer.

Wang et al. [16] discussed how JA affects the UAV's trajectory, investigating a scenario where the UAVs communicate with grounded nodes suffering a jamming signal from a stationary source. They formulated this challenge as an optimization problem, designing a trajectory planning method to optimize the drone's position in the airspace dynamically, applying the concept of slack variables. Although they addressed optimal results for a single adversary, the solution generates sub-optimal results for multiple jammers.

Wu et al. [17,18] expanded this context by exploring the QoS involved in considering the communication throughput and delay with grounded nodes, whose location guide the UAVs trajectory. They proposed two-block coordinate descent (BCD) based algorithms to solve it sub-optimally, which improved significantly different QoS requirement levels. Gao et al. [19] enhanced this model by using secondary transmitters. Duo et al. [14] proposed a similar solution but considered a scenario where drones collect data from a Wireless Sensor Network (WSN).

Some studies investigated the impact of JA in other networks integrating drones as a relay communication node, such as VANETs [20,21]. They model an anti-jamming game in which the UAV decides whether or not to relay the grounded vehicle's message to a Road Side Unit (RSU).

Recently, intelligent anti-jamming mechanisms for UAV-based networks considered machine learning-based strategies. Sedjelmaci et al. [22] proposed a reinforcement learning-based approach that handles cyber-attacks, including JA. Mowla et al. [15] designed a distributed mechanism based on Federated and Reinforcement Learning for FANETs, discussing that a distributed approach can cover several

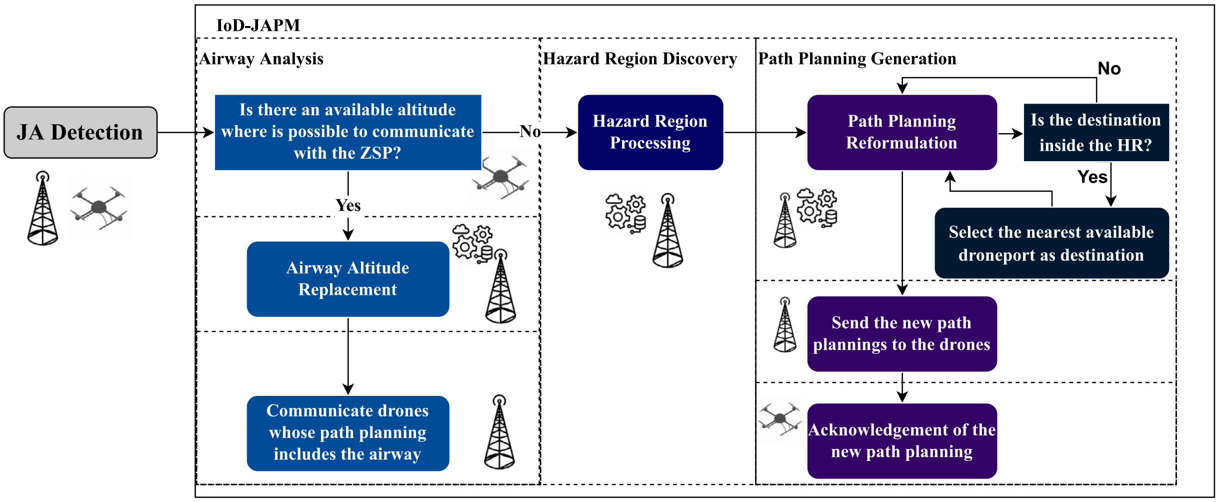


Fig. 3. The proposed IoD-JAPM workflow, given that a JA is already detected.

challenges of UAV-based networks, for instance, the Size, Weight, and Power (SWaP) limitations.

3.2. Discussion

It is noteworthy that JA has been widely investigated in the UAV context. On the one hand, all these studies assume that drones can fly freely over airspace. On the other hand, drones have limited airspace to fly in IoD. Given that a particular region is affected by JA, a drone flying over there will be under attack. Considering the airway boundaries, the drone may not have a possible trajectory to avoid the attack regardless of the strategy adopted.

Fig. 2 illustrates an example of a grounded adversary performing a JA over the IoD. The blue segments represent the airways, the red region represents the range of the performed JA, and the dashed line represents the drone trajectory. Considering the current solutions, for instance, the trajectory optimization proposed by Wu et al. [18], the drone trajectory will deviate, aiming to optimize its transmit power and the minimum delay considering the initial path planning. As we note, this deviation (orange dashed lines) can lead the drone to invade other airways not initially in its flight plan. Also, the deviation can lead to extra power consumption. These aspects pose severe risks to the availability and integrity of other drones since both collisions and power issues can occur, representing open challenges.

Our previous study [27] discussed these challenges and introduced a protection mechanism to mitigate the effects of JA over IoD. Although the proposed mechanism protects from JA, some challenges arose. Restricted airways topology (i.e., few parallel altitudes and limited flyable paths) is more affected by JA regarding the reformulation of the drone path planning and, therefore, the increasing time to complete a trip. Furthermore, the mechanism cannot correctly mitigate the effects of JA when the drone destination is inside an affected region.

Given this perspective, we propose an enhanced mechanism to mitigate the JA in the IoD environment, named IoD-JAPM. In a nutshell, the mechanism aims to detect an airway altitude free from JA, considering the LoS communication. Regarding the drones with a final destination region affected by JA, the mechanism try to allocate drones in the nearest *vertiports* until the source of JA can be countered. Hence, we advance the state-of-the-art of protection mechanisms on the IoD.

4. IoD-JAPM

In this section, we describe the design of IoD-JAPM. It aims to prevent IoD nodes suffer from JAs previously detected in the network. IoD-JAPM allows drones and ZSPs to take countermeasures to avoid the attack, ranging from analyzing the airway's availability to the potential reformulation of the drone path planning.

Fig. 3 illustrates the main steps of IoD-JAPM. Our approach is a distributed system in which drones and ZSPs operate cooperatively to avoid an ongoing JA. This attack is previously identified by some well-known jamming detection techniques [12]. As these detection techniques perform properly for UAV-based networks regardless of the airways topology, IoD-JAPM is not limited nor dependent on these methods.

IoD-JAPM has three main steps:

▶ **Airway Analysis**: in the first step, a drone in the region affected by a JA stops and elevates its altitude aiming to find a re-connection with a ZSP. If the drone can re-establish communication, there is an available altitude at which JA can be avoided. This approach is based on the available aerial space between two parallel airways and the LoS communication between the ZSPs and the drones, which are particular characteristics of the IoD. We formally describe this step in Section 4.2.

▶ **Hazard Region (HR) Discovery**: if the communication cannot be re-established, we move to the second step, where the IoD cloud system must discover all the affected areas, representing the Hazard Region. We formally describe this step in Section 4.3.

▶ **Path Planning Generation**: Once the HR is discovered, the IoD cloud system generates novel path plannings for the affected drones, avoiding the HR. However, this reformulation cannot be addressed when the drone destination is inside the HR. In this case, IoD-JAPM selects the nearest available *vertiport* as a new destination, providing a "waiting area" for the affected drone. We formally describe this step in Section 4.4.

4.1. System and adversary model

Before defining the IoD-JAPM in detail, we formalize the main elements involved in the IoD environment. They are defined as follows.

▶ $\mathcal{D}$ is the set of drones that fly over the airways;
▶ $\mathcal{P}$ is the set of path plannings defined for each $d \in \mathcal{D}$;
▶ $\mathcal{Z}$ is the set of ZSPs that manages the IoD airspace;
▶ $\mathcal{V}$ is the set of *vertiports* over the environment;
▶ λ is the IoD communication channel;
▶ $\mathcal{A}$ is the set of grounded adversaries spread over the environment;
▶ τ is an interference signal over the communication channel, representing the JA properly. This signal stems from a subset $\mathcal{A}' \subset \mathcal{A}$.

As discussed, we propose IoD-JAPM to overcome the lack of current solutions for the IoD environment, focusing on the drone trajectory through the airways. Therefore, our focus is to design a mechanism

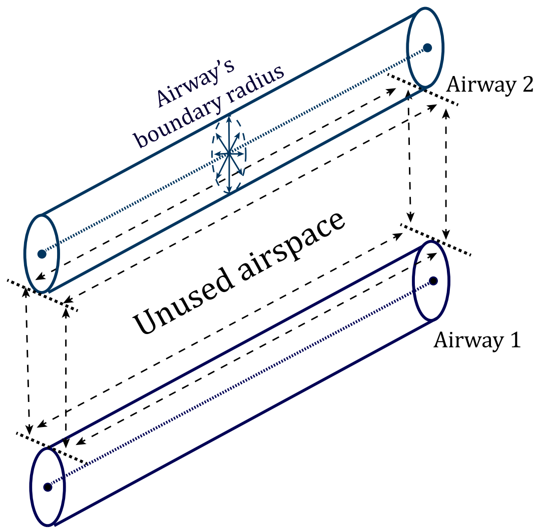


Fig. 4. The delimitation of the unused airway space between two parallel airways with different altitudes.

that allows the network to recover promptly when an attack is previously detected. Thus, we consider an IoD scenario following these assumptions:

▶ IoD nodes can detect a JA's occurrence following some detection strategies found in the literature [10,12,15,29,30]. Our mechanism is not limited nor dependent on these detection strategies, being applied one step forward when the attack is already detected;

▶ A given adversary will always perform the attack through the best configuration, considering his goals. The adversary can perform a continuous JA regardless of the power consumption of the jamming device. Furthermore, the JA can be driven either to affect the communication between the nodes or to the GPS signals;

▶ Once a given drone is affected by the attack, it can lose communication with the ZSP. In this case, the drone generally hovers or returns to the last point of a valid communication. If any anti-jamming strategy is taken, the network facilitates other attacks, such as a purposeful hijacking;

4.2. Airway analysis

As presented in Fig. 3, once the network detects the occurrence of a JA, the IoD-JAPM first step consists of a drone verifying if it is possible to establish communication with a ZSP node z still inside the airway in the affected region. Considering that in our system model the attack occurs from the ground, τ strength decreases as the drone altitude increases.

Hence, the main idea in this step is to move the drone to a higher altitude aiming to re-establish communication with a ZSP z. However, the airways have a constrained region to flight, including a maximum altitude. Nonetheless, once an airway segment β is placed, other segments should not intersect β in lower or higher altitudes unless through the insertion of an intersection node [5].

An exception occurs when β has similar airway segments defined through different altitudes, as depicted in Fig. 4. Given these assumptions, a drone d flying over an affected airway segment β has a safe altitude range to fly, ranging from its current position to a maximum altitude α. This maximum altitude can be previously defined by the ZSP when a drone requests access to the airspace, considering two main aspects: a safe distance from a parallel high-altitude airway segment when it exists; and the drone Size, Weight, and Power (SWaP) constraints. Hence, the drone can try to re-establish the connection with the ZSP over unused airspace, as illustrated in Fig. 4.

Algorithm 1: Drone-Airway-Analysis

Input : $\alpha, z, \Delta t_{ack}, \Delta t_{schedule}$

1 send a `Hello` message to z and wait a time interval Δt_{ack} for an `ACK` answer
2 **if** *There is no* `ACK` *from* z **then**
3 $\quad\alpha_{current} \leftarrow$ current drone altitude
4 $\quad$**if** $\alpha_{current} \leq \alpha$ **then**
5 $\quad\quad$adjust the navigation vectors towards α
6 $\quad\quad t_{next} \leftarrow \Delta t_{schedule}+$ the current system time
7 $\quad\quad$schedule a call to `Drone-Airway-Analysis` on t_{next}
8 $\quad$**else**
9 $\quad\quad$go back to the previous checkpoint `// All attempts to communicate fail`
10 **else**
11 $\quad$send a `Free-JA` message to z informing $\alpha_{current}$
12 $\quad$adjust back the navigation vectors towards the current checkpoint

Formally, the airway analysis step embraces both the drone and ZSP systems. Hence, we define two algorithms for this step. Algorithm 1 is a drone-centered method. Algorithm 2, in turn, is the ZSP-centered procedure.

4.2.1. Drone-centered approach

Algorithm 1 formally describes the airway analysis carried out by a given drone. Fig. 5 illustrates this central concept. As input, it requires the pre-defined maximum altitude α, the information of the ZSP z, and the time intervals Δt_{ack} and $\Delta t_{schedule}$, representing an acknowledge and schedule time intervals, respectively.

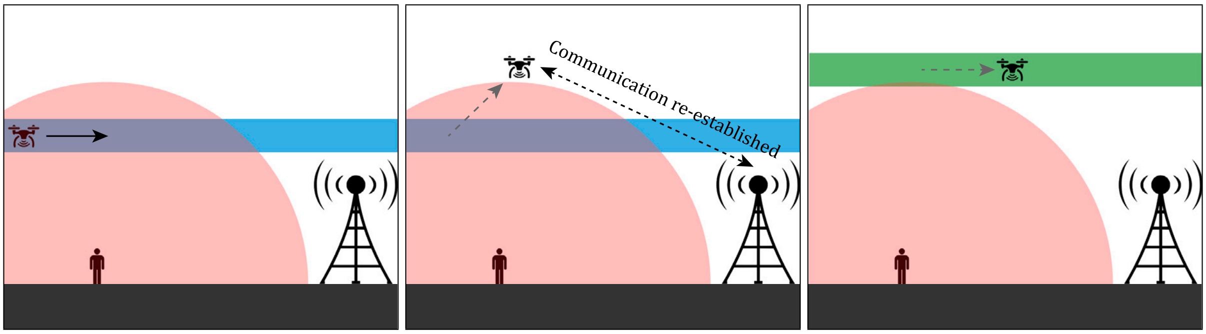
Initially, the drone tries to communicate with z, sending a `Hello` message, verifying whether the communication still works (Line 1). If, after a time interval Δt_{ack}, there is no `ACK` from z, the drone assumes that the communication is lost, initializing the communication attempts on higher altitudes (Lines 2–9), as illustrated in Fig. 5(a). Based on the current drone altitude (Line 3), it adjusts its trajectory planning if its current altitude does not reach α (Line 5). After, the drone schedules a new communication attempt based on the time interval $\Delta t_{schedule}$ (Line 6). Suppose the current altitude reaches α (Lines 8–9). In that case, all attempts fail. The ZSP will need to take additional action, isolating the affected region and reformulating the drone path planning, as described in the following subsection. In this case, the drone returns to the last checkpoint where communication occurred.

However, suppose the drone receives the `ACK` from z (Lines 11–12). In that case, the communication is re-established, indicating that a given drone flying over the current altitude $\alpha_{current}$ is not affected by the JA, as demonstrated in Fig. 5(b). In this case, the drone sends a `Free-JA` message to z, informing the current altitude and adjusting its trajectory to the present checkpoint according to its path planning.

Since z receives the new altitude, it will update this information regarding the affected airway segment and spread it to the drones whose path planning contains the affected segment. Fig. 5(c) shows this case, in which the airway with green color represents the affected airway with the new altitude. As these procedures consist of update-based messages, they are not described in detail throughout the algorithms.

▶ *Time Complexity Analysis*: in this analysis, we focus on the time complexity of the procedures inherent to the drone system. Therefore, this analysis does not cover "inter-node" procedures (Lines 1 and 11). In summary, Algorithm 1 consists of direct assignments to primitive data structures inherent to the drone system (Lines 3, 5, 6, 7, 9, 12). Hence, the time complexity of Algorithm 1 is given by Eq. (1):

$$T(\text{Drone-Airway-Analysis}) = \Theta(1) \tag{1}$$



(a) A drone inside an affected region loses its communication with the ZSP

(b) The drone elevates its altitude trying to re-establish the communication

(c) The ZSP set a new altitude to the affected airway

Fig. 5. Airway analysis step, performed by a drone.

Algorithm 2: ZSP-Check-Drone-Upd

Input : $\alpha, \mathcal{Q}_{upd}, \mathcal{D}_{JA}$

1 $d_{upd} \leftarrow$ extract element of the top from $\mathcal{Q}_{upd}$
2 **if** d_{upd} *did not update its location when expected* **then**
3   **if** $d_{upd} \notin \mathcal{D}_{JA}$ **then**
4         $\mathcal{D}_{JA} \leftarrow \mathcal{D}_{JA} \cup d_{upd}$
5         $l_{last} \leftarrow$ get last location update of d_{upd}
6         $t \leftarrow$ predict the time that d_{upd} will reach α from l_{last}
7   insert d_{upd} in $\mathcal{Q}_{upd}$ with priority t
8   schedule a call to `ZSP-Verify-Drone-Update` at t
9   **else**
10         $l_{predict} \leftarrow$ predict the d_{upd}'s location if it had updated its location
11   `Cloud-Calc-Hazard-Region(`$l_{predict}$`)`

4.2.2. ZSP-centered approach

Bearing that drones cannot re-establish communication with the ZSP even after analyzing other altitudes, a given ZSP z monitors the drone location update. Thus, the IoD cloud system can take other actions to avoid the JA (defined in the following sections). Algorithm 2 describes the ZSP monitoring. Besides the maximum altitude α, the algorithm requires as input a priority queue $\mathcal{Q}_{upd}$ and a set $\mathcal{D}_{JA}$. $\mathcal{Q}_{upd}$ stores the information of drones ordered by the next expected location update time. $\mathcal{D}_{JA}$ stores drones trying to find an altitude free from JA. The recurring drones' location update messages fed the data structures inherent to the IoD management system.

Initially, the mechanism takes d_{upd} (that would have updated its position) from $\mathcal{Q}_{upd}$ (Line 1). If d_{upd} did not update its location at the current time, it could indicate a possible communication loss due to the JA (Lines 2–11). However, it is necessary to consider the time interval the drone potentially takes to reach α.

Hence, the algorithm verifies if d_{upd} already belongs to $\mathcal{D}_{JA}$ (Line 3). The negative case corresponds to the first time d_{upd} has not updated its location. Therefore, the mechanism includes it in $\mathcal{D}_{JA}$ (Line 4) and predicts the expended time t to reach the altitude α, based on the d_{upd}'s last location update (Lines 5–6). The d_{upd} is re-inserted in $\mathcal{Q}_{upd}$, now with the priority t (Line 7). Finally, the mechanism schedules a call to this procedure at the time t, aiming to verify if a new altitude was discovered during this time (Line 8).

In the affirmative case, d_{upd} has failed to find an available altitude to re-establish the communication, in which the IoD cloud system must take additional countermeasures to avoid JA (Lines 10–11), as described in Fig. 3. Thus, the mechanism predicts the location of d_{upd} if it had updated its position, and passes this information as input to Algorithm 3, responsible for identifying the Hazard Region (HR), leading us to the next step of IoD-JAPM.

► *Time Complexity Analysis*: to conduct this analysis, we consider that the element of the top of $\mathcal{Q}_{upd}$ did not update its location when expected, representing the worst case. Also, let us consider a heap data structure for $\mathcal{Q}_{upd}$.

The extraction of the top element takes $\mathcal{O}(\log n)$ (Line 1). Considering that the drone d_{upd} does not belong to $\mathcal{D}_{JA}$ (Lines 3 to 8), both the set union and the l_{last} assignments have a constant time complexity $\mathcal{O}(1)$ (Lines 4 and 5). On the other hand, the priority queue insertion of Line 7 takes $\mathcal{O}(\log n)$, dominating this conditional. If the drone already belongs to $\mathcal{D}_{JA}$, the drone location prediction (Line 10) and the calling for Algorithm 3 also take $\mathcal{O}(1)$. Hence, the time complexity of Algorithm 2 is given by Eq. (2):

$$\begin{aligned} T(\text{ZSP-Check-Drone-Upd}) &= 2\mathcal{O}(\log n) + \mathcal{O}(1) \\ &= \mathcal{O}(\log n) \end{aligned} \tag{2}$$

4.3. Hazard region

If the drones fail to identify an altitude free from JA, the next step consists of isolating the affected region, named *Hazard Region* (HR). Considering the defined graph representation, a given location l is part of an airway line segment delimited by $\langle w_1, w_2 \rangle \in G.E$, such that the direction vector is $\vec{w_1 w_2}$. Hence, to delimit the HR, the system considers three different cases, described as follows.

1. The $\langle w_1, w_2 \rangle \in G.E$ that l belongs must be part of HR;
2. $\forall \langle u_1, u_2 \rangle \in G.E$, if $\langle u_1, u_2 \rangle$ is inside a spherical HR with radius r from the l's geographic coordinates, then it must be part of HR;
3. $\forall \langle u_1, u_2 \rangle \in \text{HR}, \forall \langle v, u_1 \rangle \in G.E$, if the node v has u_1 as the unique adjacent node, then $\langle v, u_1 \rangle$ must be part of HR. In other words, airway segments that reach an already affected segment and have no other airway segment to follow also must be part of HR;

Algorithm 3 defines these cases. It gives as output a subgraph $G_{HR} \subset G$ that represents the HR. Initially, G_{HR} is initialized empty (Line 1). After, all G edges are visited (Lines 2–6) to verify whether the airway segment is close enough to l considering the radius r. In the affirmative case, the segment is included in the HR (Lines 4–6). As the distance between l and its airway is near 0 (and less than r), this step considers cases (1) and (2). With the initial HR established, the system focuses on verifying if there are remained airway segments that have no other airway segment to follow — case (3). Thus, we define a subgraph $G_{rmnd} \subset G$ whose edges do not already belong to HR (Line 7). If there is an airway segment $\langle u, v \rangle \in G_{rmnd}$ that goes to the HR with any other airway to follow, it reaches a node $v \in G_{rmnd}.V$ whose outdegree is 0. It is always true since the affected airway segments are in G_{HR}.

Another case can occur when the IoD airway topology has sink nodes that represent, for instance, a drone company garage. To avoid this case, it is necessary to verify if the node $v \in G_{HR}$ (Lines 8–13). When these conditions are satisfied, every airway segment that goes to

Algorithm 3: Cloud-Calc-HR

Input : l
Output: G_{HR}
1 $G_{HR} \leftarrow \emptyset$
2 **foreach** $\langle w_1, w_2 \rangle \in G.E$ **do**
3 $dist \leftarrow$ calculate the distance from l to the line segment $\langle w_1, w_2 \rangle$
4 **if** $dist \leq r$ **then**
5 $G_{HR}.V \leftarrow G_{HR}.V \cup \{w_1, w_2\}$
6 $G_{HR}.E \leftarrow G_{HR}.E \cup \{\langle w_1, w_2 \rangle\}$
7 $G_{rmnd} \leftarrow G - G_{HR}$
8 **while** $(\exists v \in G_{rmnd}.V, outdegree(v) = 0) \land (v \in G_{HR})$ **do**
9 **while** $\exists \langle u, v \rangle \in G_{rmnd}.E$ **do**
10 $G_{HR}.E \leftarrow G_{HR}.E \cup \{\langle u, v \rangle\}$
11 $G_{rmnd}.E \leftarrow G_{rmnd}.E - \{\langle u, v \rangle\}$
12 $G_{HR}.V \leftarrow G_{HR}.V \cup \{v\}$
13 $G_{rmnd}.V \leftarrow G_{rmnd}.V - \{v\}$
14 Cloud-Reformulate-PP(G_{HR})

v is added to G_{HR} and removed from G_{rmnd} (Lines 9–11). At the end of this iteration, the incidence list of v will be empty. Thus, v is added to G_{HR} and removed from G_{rmnd} (Lines 12–13). When there are no more nodes with outdegree equals 0, the HR is completely discovered and can be used to identify the drones needing a path planning reformulation (Line 14).

▶ *Time Complexity Analysis*: the initialization of the HR as an empty set (Line 1) has $\Theta(1)$ complexity. The loop from Line 2 to 7 iterates over all edges of G, performing constant mathematical operations (Line 3) and set operations (Lines 5 and 6). All of them have $\mathcal{O}(1)$ complexity. Hence, the loop is $\Theta(|G.E|)$. The deployment of the remained graph G_{rmnd} occurs through set operations (Line 7), i.e., it is $\Theta(1)$.

The remained part of the algorithm consists of two nested loops (Lines 8 to 13). Considering an aggregate analysis, these loops will iterate over the edges, in the worst case. Hence, the complexity is $\mathcal{O}(|G.E|)$. Lastly, a call to Algorithm 4 is described in the next section. Eq. (3) gives the time complexity of Algorithm 3:

$$\begin{aligned} T(\text{Cloud-Calc-HR}) &= \Theta(1) + \Theta(|G.E|) + \mathcal{O}(|G.E|) \\ &= \Theta(|G.E|) \end{aligned} \tag{3}$$

4.4. Path planning reformulation

Algorithm 4 describes the reformulation of the path planning performed in the IoD cloud system. It consists of identifying if, given a path planning $p \in \mathcal{P}$, there are airways into the HR that the drone $p.d$ will fly until the end of its trip. In the affirmative case, the system generates a new one and communicates it to the drone. Furthermore, if the drone destination point is not reachable anymore, the system tries to redirect the destination to a *vertiport* until the jamming source is actively countered.

As input, it receives the HR graph representation provided by Algorithm 3. To reformulate the path planning, we calculate first what airways are not affected by the HR (Line 1). Then, the mechanism verifies each path planning $p \in \mathcal{P}$ to identify if it will demand a reformulation. There are two auxiliary nodes, u and v, representing the last geographic point before the current path enters HR and the first geographic point after leaving HR. They are both initialized as null pointers (Line 3). Also, we initialize as *false* a flag regarding the need to designate a *vertiport* as a destination point to the drone, in the case it is not possible to generate a new path (Line 4).

Similarly, an auxiliary path p_{aux} records the reformulated path planning. It is initialized with the same attributes as p except the path planning, which is given until the current airway that the drone is flying, indicated by the *order* attribute (Line 5). After, IoD-JAPM verifies each airway segment $e \in p.G'.E'$ from the current drone airway to the last one (Lines 6–22). If the verified e is in the HR (Lines 7–9), it is necessary to check if it is the first segment inside the HR, which means that u points to null. In this case, u receives the origin node of e (Lines 8–9).

When e does not belong to HR (Lines 10–22), we need to consider two different situations regarding the previous edges: they are outside the HR, which means that e follows a “free HR path”; or part of them is in the HR. In the first case, e is added to the path in p_{aux} (Lines 11–12). Otherwise, the origin point $e.w_1$ is the first point outside HR associated with v (Line 14). Therefore, given points u and v, the minimum path planning over G_{avail} is calculated (Line 15). If there is an available path E_{new}, it is added to the new course (Lines 16–18). As an HR segment is processed and replaced, both u and v are set to null (Line 19). However, if the algorithm does not return any path, there is no alternative path to the drone. Thus, the system must process if there is an available *vertiport* where the drone can hold on until the jamming source is countered actively (Lines 20–21). After processing the airway segments of p, the system must certify if the last segment is not inside the HR. It can be verified through u, which must be null. If the segment is inside the HR (Lines 23–24), there is no alternative path to finish the trip. Likewise, in the case of Lines 20–21, it is necessary to evaluate if there is an available *vertiport*.

If one of the two above conditions occurs, the algorithm chooses the best *vertiport* for the drone (Lines 25–40). Summarily, we calculate the shortest path from the drone to each *vertiport* (since it has a “slot” to shelter the drone), selecting the one with the shortest distance (Lines 26–38). If there is no available *vertiport*, the cloud system emits a message to the drone informing that there is no available path to fly (Line 40). In this case, the drone service provider must take an external action unrelated to our mechanism.

Finally, if the path was affected in one or more segments, the current course is replaced by the new one and submitted to the nearest ZSP to $p.d$, informing the drone about the update (Lines 41–43).

▶ *Time Complexity Analysis*: this algorithm has a series of simple assignments and variable initializations (Lines 3, 4, 5, 9, 12, 14, 17–19, 21, 24, 26–29, 32, 35–36, 38, 42), we state all of them as $\mathcal{O}(1)$. The main part of Algorithm 4 iterates over the defined $|d|$ path plannings (Lines 2–40). This iteration has an inner loop going through all the edges of the path planning in the worst case (Lines 6–22). Also, the size of edges in the path planning is limited to $|G.E|$, although it hardly happens in practical terms. Inside this inner loop, the Dijkstra call of Line 15 dominates the entire time complexity of the loop in the worst case. Hence, the inner loop has a time complexity of $\mathcal{O}(|G.E|(|G.V| + |G.E|\log|G.V|))$. As the airways topology of IoD is always a connected graph [5], $|G.E| \geq |G.V|$. Therefore, this time complexity can be reduced to $\mathcal{O}(|G.E|^2 \log|G.V|)$.

Besides this inner loop, the *vertiport* processing (Lines 25–40) has significant time complexity. This task iterates over all the *vertiports*, which is limited by $|G.V|$. Nonetheless, it hardly happens in real scenarios since it would mean that all the available points would be *vertipoints*. In this iteration, another Dijkstra algorithm dominates the time complexity. Hence, the iteration has a time complexity of $\mathcal{O}(|G.V|(|G.V| + |G.E|\log|G.V|)) = \mathcal{O}(|G.V|(|G.E|\log|G.V|))$.

Given this discussion, Eq. (4) presents the time complexity of Algorithm 4:

$$\begin{aligned} T(\text{Cloud-Rfrmlt-PP}) &= |d|(|G.E|^2 \log|G.V| \\ &\quad + |G.V|(|G.E|\log|G.V|)) \\ &= |d|(|G.E|^2 \log|G.V|) \end{aligned} \tag{4}$$

Although Algorithm 4 presents a theoretical quadratic time complexity, it does not occur in practice since the path planning has several edges significantly less than the whole topological graph.

Algorithm 4: Cloud-Rfrmlt-PP

Input : G_{HR}
$G_{avail} \leftarrow G - G_{HR}$
foreach $p \in \mathcal{P}$ **do**
  $u, v \leftarrow null$
  needsVertiport $\leftarrow$ *false*
  $p_{aux} \leftarrow \langle p.d, p.G'.E'_{\langle 0..p.airway.order\rangle}, p.airway\rangle$
  for $e \in p.G'.E'$ such that $p.airway.order \leq e.order < |p.G'.E'|$ **do**
    if $e \in G_{HR}.E$ **then**
      if $u = null$ **then**
        $u \leftarrow e.w_1$
    else
      if $u = null$ **then**
        $p_{aux}.G'.E' \leftarrow p_{aux}.G'.E' \cup e$
      else
        $v \leftarrow e.w_1$
        $dist,\ E_{new} \leftarrow$ Dijkstra(G_{avail}, u, v)
        if $E_{new} \neq \emptyset$ **then**
          $p_{aux}.G'.E' \leftarrow p_{aux}.G'.E' \cup E_{new}$
          $p_{aux}.G'.E' \leftarrow p_{aux}.G'.E' \cup e$
          $u, v \leftarrow null$
        else
          needsVertiport $\leftarrow$ *true*
          break loop
  if $u \neq null$ **then**
    needsVertiport $\leftarrow$ *true*
  if needsVertiport **then**
    source $\leftarrow$ last node of $p_{aux}.G'.E'$
    closest $\leftarrow$ null
    $path_{best} \leftarrow$ null
    $dist_{short} \leftarrow \infty$
    foreach $vp \in \mathcal{V}$ **do**
      if vp **is not** full **then**
        $l_{vp} \leftarrow vp$ location
        $dist, path \leftarrow$ Dijkstra(G_{avail}, $source$, l_{vp})
        if $dist < dist_{short}$ **then**
          $closest \leftarrow vp$
          $dist_{short} \leftarrow dist$
    if $closest \neq$ null **then**
      $p_{aux}.G'.E' \leftarrow p_{aux}.G'.E' \cup path_{best}$
    else
      send a message to $p.d$ informing that there is no available path to fly
  if $p_{aux}.G'.E' - p.G'.E' \neq \emptyset$ **then**
    $p \leftarrow p_{aux}$
    ZSP-Drone-PP-Updt(p)

5. Experimental evaluation

The main goal of this work is to analyze how JA impacts drone path planning in IoD. As discussed earlier, our previous protection mechanism provides adequate protection against the JA [27], however, the drone trajectory is affected strongly, leading us to propose the IoD-JAPM. Therefore, our evaluation focuses on both trajectory and path planning rather than the avoidance of JA.

We conducted an experimental evaluation through simulations to investigate the following aspects: (i) how JA affects the drone mobility in the IoD environment; and (ii) how IoD-JAPM contributes to overcoming the challenges introduced by our previous study [27]. In this section, we present the evaluated scenario, the simulation parameters, and the metrics.

5.1. Compared approaches

We consider four approaches to evaluate the proposed mechanism. We divide them into a scenario without JA (representing an optimum regarding the drone trajectory) and three scenarios where different solutions mitigate the JA.

5.1.1. Free-JA

In this approach, there is no JA underway. Drones will communicate and move as much as possible following the airways. This approach represents an "optimum" scenario in terms of flight time and power consumption.

5.1.2. Baseline solution [27]

This approach focuses on our previous mechanism [27] to detect and mitigate a long-term JA, performed by a malicious entity on the ground, keeping stationary mobility.

5.1.3. Joint optimization approach [18]

An important aspect when analyzing the robustness of IoDJAPM is to compare our solution with a state-of-the-art method. However, these methods consider the airspace as free-to-fly (as discussed in Section 3.1), differing from our assumptions. Thus, we need to consider an approach as similar as possible to evaluate a fair comparison. Bearing this in mind, we consider the approach proposed by Wu et al. [18], denoted as "Joint". The proposed method aims to optimize jointly the communication throughput and the UAV's mobility. The UAVs also move towards well-defined locations: the targeted grounded nodes. Considering our proposed scenario, they can be taken as terrestrial waypoints of the UAV's trajectory.

Nonetheless, the main difference between the approaches is airspace navigability. Hence, we adapt the proposal of Wu et al. [18] so that the drone trajectory remains inside the airway's boundary radius and, in a JA situation, the "unused airspace" (as described in Fig. 4).

We define new constraints regarding the mobility capability of the UAV to meet this adaptation, presented in Eqs. (5)–(7).

$$\mathtt{Min}_{dist}(\mathrm{q}_u[i], \beta) \leq r \tag{5}$$

$$\mathrm{q}_u[i]^z \leq \alpha \tag{6}$$

$$\mathtt{distance}(\mathrm{q}_u[i], w) \leq \mathtt{distance}(\mathrm{q}_u[i-1], w) \tag{7}$$

In Eq. (5) we define that the minimum distance between the position q_u of a given drone u at the moment i and a given line segment of an airway β must be less or equal to a radius r. In other words, the calculated position must be inside the boundaries of the airway β. In Eq. (6) we define that the calculated altitude (corresponding to the z-axis on the Euclidean system) must be less or equal to the maximum available altitude α.

Since the proposal [18] aims to jointly optimize the drone trajectory and the throughput with a targeted node, the calculated positions tend to approximate drones to the ZSP instead of the defined waypoint. To avoid this case, we define in Eq. (7) that the distance between the calculated position of the drone and the waypoint w must be less than the previous distance. Thus, we ensure that the drone moves towards the waypoint instead of the ZSP. All these constraints are valid and applicable to all the assumptions, equations, and algorithms presented by Wu et al. [18].

5.1.4. IoDJAPM

This approach is similar to **Baseline**, but the network operates with IoD-JAPM to mitigate the JA impact. Hence, it is possible to compare the performance of **IoDJAPM** with both the former approaches, namely **Baseline** [27] and **Joint** [18] methods.

5.2. Evaluated scenario

We consider two distinct airway topologies, $\mathbf{T}_1$ and $\mathbf{T}_2$, summarized in Table 1. T_1 is a robust infrastructure where drones have parallel airways and different flying paths, designed following a piece of the roadside's structure in Manhattan Island, NY. In turn, T_2 is a constrained infrastructure, where drones have limited paths to fly at a single available altitude. This topology follows a piece of the roadside structure of San Francisco, CA. Also, T_1 has two available *vertiports* with a double capacity as the single *vertiport* of T_2.

5.3. Simulation parameters

We performed the simulations using the IoDSim,[1] an OMNET++ integrated framework designed for IoD. Table 2 shows a list of simulation parameters, mainly regarding the environmental nodes. These parameters are defined considering a realistic scenario with a reasonable time for an attacker to perform JA [31]. Likewise, the distributed system can detect and act against the attack.

Table 1
Topology attributes.

Attr.	$\mathbf{T}_1$	$\mathbf{T}_2$
Region size	$3 \times 3 \times 0.3\,\mathrm{km}^3$	$4 \times 2 \times 0.1\,\mathrm{km}^3$
#available altitudes	3	1
#Airway nodes (total)	402	60
#Airway segments (total)	997	98
#*Vertiports*	2	1
Vertiport capacity	4	2

Table 2
Simulation parameters.

Parameter	Value
Scenarios	**Free, Baseline, Joint, IoDJAPM**
#Jammers	**{1,2}**
Topologies	$\{\mathbf{T}_1, \mathbf{T}_2\}$
Simulation time	20 min
Drone speed	Uniform [5–10] m/s
Mobility pattern	Gauss–Markov ($\alpha = 0.7$)
#Drones	25
Jammer's mobility	Grounded, Stationary
Jammer's behavior	Long-term attack
#ZSPs	4
ZSP update check interval	1 s
ZSP Update check threshold	5
Transport protocol	UDP
Routing protocol	AODV
MAC protocol	CSMA/CA
Radio's modulation	APSK
Radio's frequency	2.4 GHz
Radio's bandwidth	2 MHz
Radio's transmission power	220 mW
SNIR threshold	4 dB
Convergence threshold μ (Joint [18])	10^{-3}

We evaluate the scenarios with one and two jammers. In both cases, the first jammer has a low interference signal range, affecting only the airway at the lower altitude. The second jammer, in turn, has a wide interference range, performing a JA over all the parallel airways of an affected region.

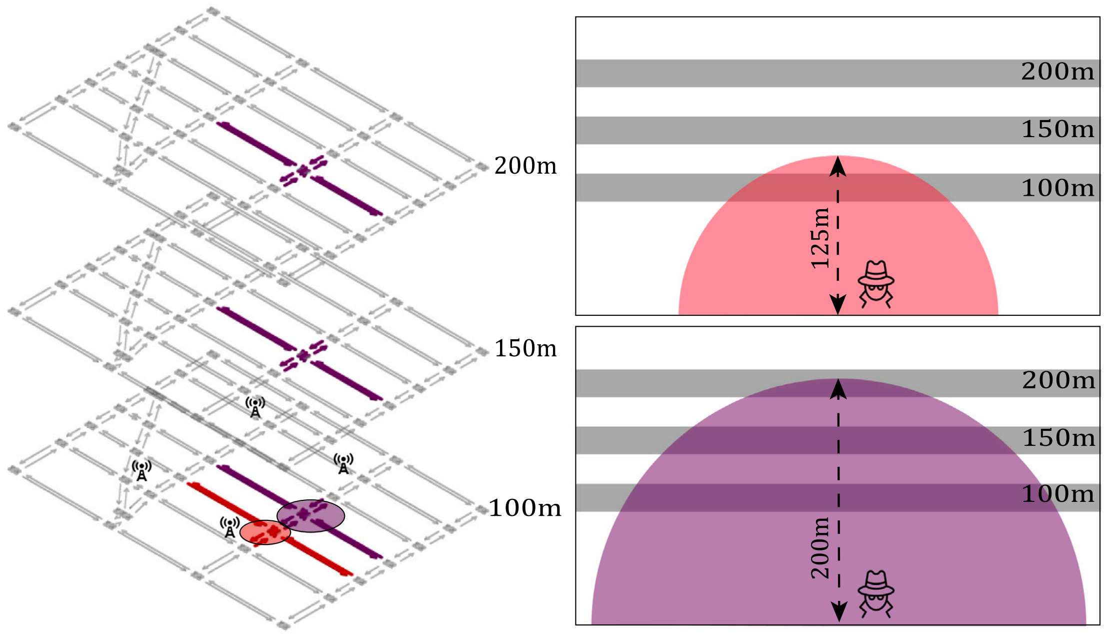
Each simulation has 20 min. We set 25 drones following a Gauss–Markov mobility pattern with speeds varying from 5 to 10 m/s. The Gauss–Markov model is configured with an angle $\alpha = 0.75$ over the navigation vectors. Thus, a given drone flying towards an aerial waypoint has small deviations in its flight, simulating potential weather conditions in the environment, such as the wind performance. We spread four ZSPs on the ground communicating with both the drones and the cloud system. Regarding communication, the protocols and radio configuration are the same for all nodes. The designed framework considers UDP, AODV, and CSMA/CA as the transport, routing, and MAC protocols, respectively. The radio has an APSK modulation, a frequency of 2.4 GHz, and a transmission power of 220 mW. The SNIR threshold is set as 4 dB.

For each combination of scenario × topology × number of jammers (except for Free-JA with Jammers), we conducted 35 experiments with distinct seeds and path planning of drones, leading to results with a confidence interval (CI) of 95%. The Jammer and ZSPs positions were the same for all replications. Hence, each experiment reflects a different drone mobility scenario over the same attackers' positions.

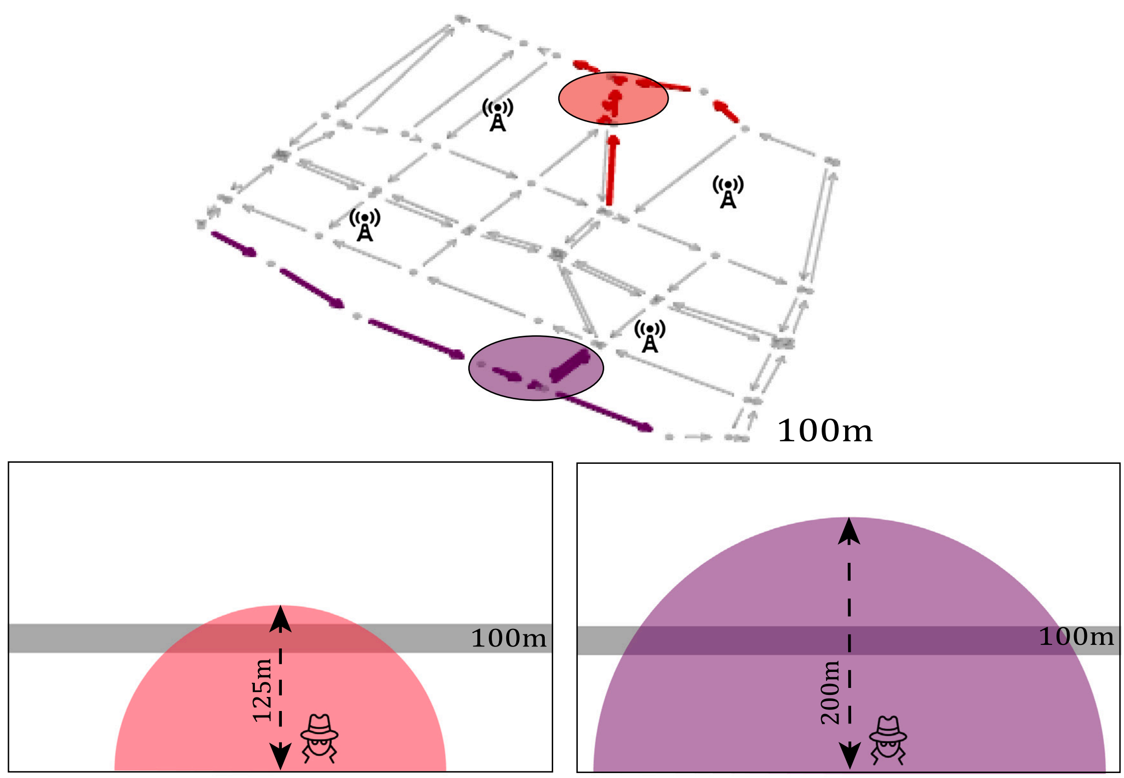
5.4. Metrics

We consider four metrics to measure how JA affects the drone path planning, focusing on the topological airways affected by the JA, the number and type of affected planning, the increasing flight distance, and the increasing power consumption. They are described as follows.

[1] Available at https://iodsim.manna.team.



(a) Hazard Region of the topology $\mathbf{T}_1$



(b) Hazard Region of the topology $\mathbf{T}_2$

Fig. 6. The typical Hazard Region calculated in each topology over all the airways. A red segment indicates an airway segment compromised by the first jammer while the purple segment refers to the second jammer.

5.4.1. Hazard Region Rate (HRR)

HRR measures how large the HR is compared to the topological graph. Eq. (8) formally describes how to calculate *HRR*, where G_{HR} is the graph corresponding to the HR and G is the topological airway graph of the network.

$$HRR = \frac{|G_{HR}.E|}{|G.E|} \tag{8}$$

5.4.2. Drones with Affected Path Planning Rate (DAPPR)

This metric indicates the rate of drones whose path planning was affected at least once due to JA compared to the total number of drones. We consider affected path plannings those routes that need to be reformulated, redirected to a *vertiport*, or those that could not proceed to a final destination. Hence, it does not embrace the flights that could proceed after a successful airway analysis. Formally, Eq. (9)

defines *DAPPR*. $|\mathcal{D}_{affected}|$ represents the total number of drones with an affected path planning, such that the ones reformulated, redirected to a *vertiport*, or that could not proceed to the final destination.

$$DAPPR = \frac{|\mathcal{D}_{affected}|}{|\mathcal{D}|} \tag{9}$$

5.4.3. Increasing Flight Distance Rate (IFDR)

IFDR calculates the relationship between the total distance between the Free-JA scenario (original) and the corresponding path planning in the scenarios with an anti-jamming mechanism. Eq. (10) defines *IFDR*, where PP_o is the original path, PP_c is the corresponding path in the Baseline or IoD-JAPM scenario, and $dist()$ is a function that calculates the total distance of a given path.

$$IFDR = \frac{dist(PP_c)}{dist(PP_o)} \tag{10}$$

5.4.4. Increasing Power Consumption Rate (IPCR)

Similar to *IFDR*, this metric compares the relation between the original and the corresponding reformulated path planning in terms of power consumption. Eq. (11) describes *IPCR*, where $\mathcal{E}_o$ is the power consumption of the drone considering the Free-JA scenario, and $\mathcal{E}_r$ is the power consumption of the corresponding drone with the reformulated path planning.

$$IPCR = \frac{\mathcal{E}_r}{\mathcal{E}_o} \tag{11}$$

6. Results and analysis

This section presents the results obtained through the simulations. We discuss each metric's results thoroughly, describing how IoD-JAPM mitigates the impact of JA, promoting a safer IoD environment. For all results, the interval errors are less than 1 measure unit. Therefore, they are not represented in the charts.

6.1. HRR analysis

Understanding the dimension and how HR affects the airspace is the first door to analyzing the impact of JA over the IoD. Fig. 6 shows the calculated HR for T_1 (Fig. 6(a)) and T_2 (Fig. 6(b)), considering the region of the two jammers. Regarding Fig. 6(a), the overlapped airways are out of scale compared to the other elements to facilitate the visualization. Also, the grounded elements (ZSPs and the region over the JA) are represented together with the lower airway.

Elements with red color refer to the first jammer (the one with less “jamming power”), and the purple color refers to the second jammer. Hence, the red circle indicates the region over the active JA from the first jammer, while the purple circle refers to the JA from the second jammer. Considering that these circles illustrate only a slice of the jammed region on the illustration plan, we provide a vertical visualization of these regions, which explains why the second jammer can affect all three altitudes of T_1. Likewise, the red and purple airway segments correspond to the HR calculated following Algorithm 3.

Fig. 7 presents the results of *HRR*, giving a numerical meaning to the topological representations. The lower the *HRR*, the lower the impact on the IoD environment. The robustness of the topology is a determining factor in the HR coverage over the environment. Even reaching the three parallel altitudes, the HR caused by the two jammers acting together over T_1 is less than the HR over T_2 caused by a single jammer. Indeed, the parallel airways of T_1 provide a drone with several ways to avoid JA, as reflected in the *HRR*.

Furthermore, even though the HRR of T_2 does not appear to be such a high rate initially, it significantly restricts the airspace for drones, impacting their path planning directly. In the following sections, we discuss these aspects in detail.

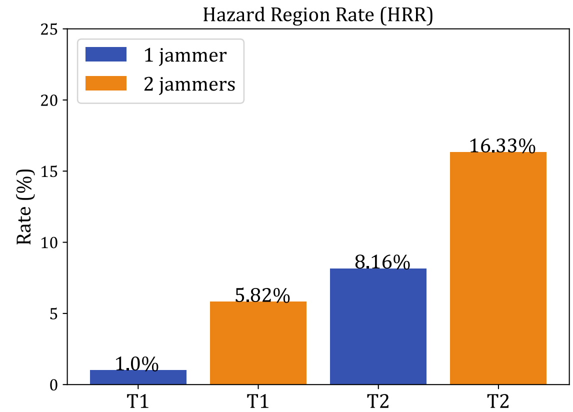


Fig. 7. Results of *HRR* for each topology and number of jammers.

6.2. DAPPR analysis

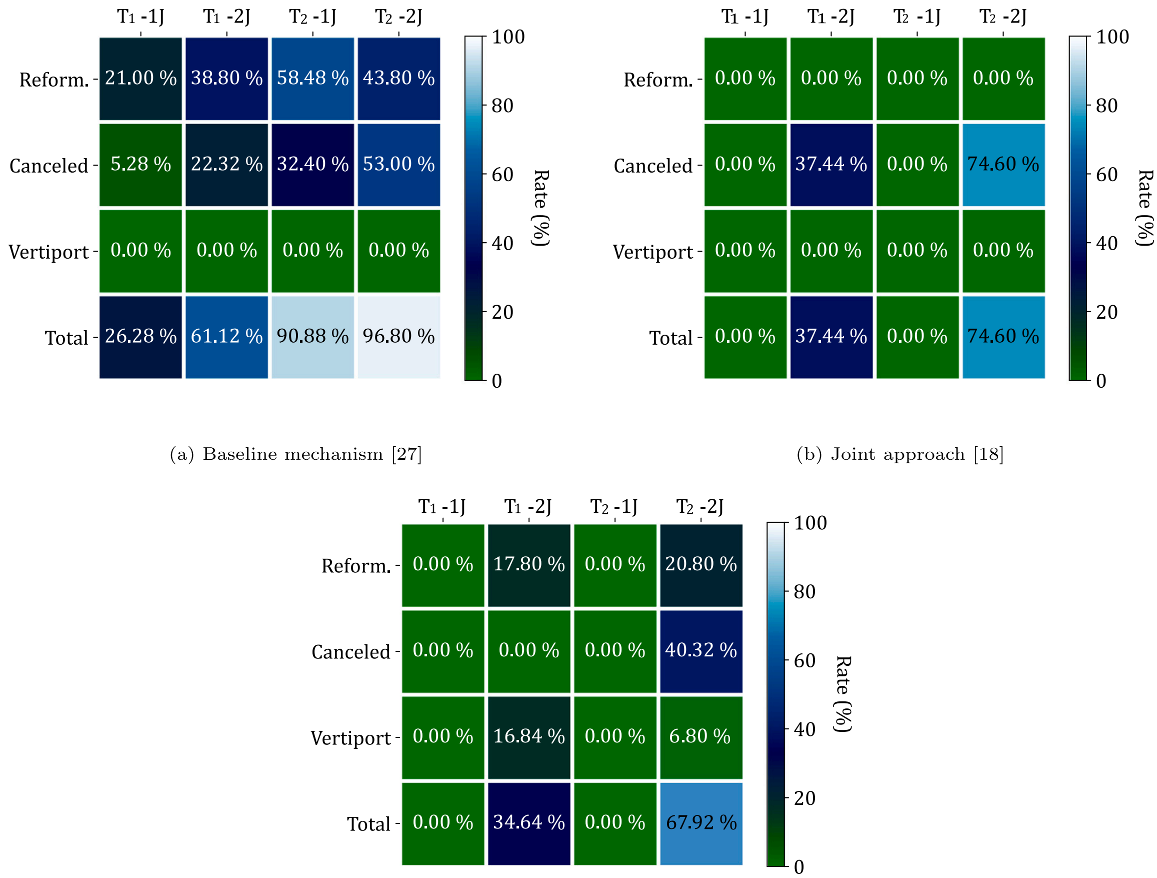
The *DAPPR* metric gives a profile of the trajectories in each evaluated configuration. Fig. 8 presents these results for the evaluated approaches: our previous solution [27] in Fig. 8(a), the Joint approach [18] in Fig. 8(b), and IoD-JAPM in Fig. 8(c). Each cell represents the *DAPPR* of a given type of path planning (lines) affected in a given configuration (columns). The last line presents the sum of the rates of the configuration rates: the less the total *DAPPR*, the less the impact of *JA* over the IoD environment.

The total rate of affected path planning highlights that IoD-JAPM can mitigate the impact of JA at a higher level than the baseline [27] and Joint [18] approaches. For the scenarios with one jammer, IoD-JAPM prevented all drones from being attacked during all experiments. Compared with the baseline mechanism, the rate is higher than 90% on average for the topology T_2. These aspects reveal how vital the altitude analysis step is, even considering a restricted topology.

Observing the results related to the presence of two jammers, IoD-JAPM overcame the Joint approach [18], especially in Topology T_2. Nonetheless, Fig. 8(b) shows the limitations of the Joint approach [18] in a restricted scenario with well-defined airways. In this case, the flights were canceled because the drone position must remain in a range based on the drone maximum speed. Therefore, the mechanism cannot find a possible trajectory without violating the imposed constraints [18]. Indeed, for the topology T_2, with two jammers, almost 75% of the drones suffered from this issue, on average. As presented in Fig. 6(b), the jamming range of the second jammer in T_2 is stronger than the first one, which can explain the obtained results.

Also, the use of *vertiports* mitigates the impact of JA, avoiding the cancellation of a significant portion of path planning compared to the baseline. For the topology T_1, it was not necessary to cancel any path planning, while for T_2 there was a decrease of 25% from the corresponding scenario with the baseline mechanism. This aspect is strictly related to the robustness of the topology. While in T_1, there are two available *vertiports* (which can assist four drones each), T_2 has only one *vertiport* (assisting two drones only). Indeed, in most experiments regarding T_2 with two jammers, the *vertiport* was full in most experiments.

Therefore, these results reveal a fundamental challenge: even with IoD-JAPM showing adequate protection against JA, the topology of the airways is still a risk factor to be considered in IoD. When such a topology has few available paths for the drone flight and does not offer a reasonable number of available flight altitudes, attackers with greater interference power can significantly impact the drone trajectory and, consequently, the provided service.



(a) Baseline mechanism [27]

(b) Joint approach [18]

(c) IoD-JAPM

Fig. 8. Results of *DAPPR* regarding the baseline mechanism [27], Joint approach [18], and IoD-JAPM.

6.3. IFDR analysis

As we discussed previously, drones have SWaP limitations, being essential to assess factors that can lead to the excessive expenditure of these resources. One of these factors is the increasing flight distance caused by applying the anti-jamming mechanisms measured by the *IFDR*. The less the *IFDR*, the less the impact on the IoD. Fig. 9 summarizes the obtained results regarding this metric.

IoD-JAPM overcomes the baseline mechanism [27] in all scenarios. Indeed, IoD-JAPM has an increase significantly less than the baseline and a similar rate compared to the Joint approach in scenarios with one jammer. In these cases, the slight increase occurs due to the altitude analysis step of IoD-JAPM, in which the drone performs minor deviations from its original path, as presented in Fig. 5. As there is no other reformulation, this is the only kind of increase.

Analyzing the results of Topology T_2 with two jammers, IoD-JAPM increased twice the flight distance compared to the Joint approach [18]. This behavior occurs mainly because there is no reformulation of path planning on the Joint approach. Therefore, the trajectory deviations are restricted to the maneuvers performed by the drones to optimize the throughput with the ZSP communication. Indeed, if we analyze this result allied with the DAPPR (Fig. 8), IoD-JAPM still is the best mechanism against JA since it cancels fewer flights than the Joint approach, reformulating a considerable amount of them as well as sending drones to vertiports.

Still considering the results of *DAPPR*, the altitude analysis performed by IoD-JAPM explains the better results obtained. The rate of reformulated path planning is always less than the baseline mechanism. Even so, some path plannings reformulated by the baseline mechanism did not change when IoD-JAPM was applied but proceeded normally because the altitude analysis was successful.

Furthermore, we can note an outlier in the results of T_2 of the baseline mechanism: the *IFDR* of one jammer was higher than two jammers, going in the opposite direction than expected. Analyzing the results carefully, we note that it occurs due to the decrease of reformulated path plannings from one jammer to the scenario with two jammers. A significant portion of flights was canceled when two jammers were in the environment, as presented in Fig. 8(a). Some of these flights were reformulated in the presence of a single jammer. This reformulation allows the drone to cross the entire environment since the second jammer region is on the opposite side. Therefore, these trajectories contribute significantly to the increasing distance, but they do not appear in the scenario with two jammers since they are canceled.

6.4. IPCR analysis

Fig. 10 presents the results of *IPCR*. The less the *IPCR*, the less the impact on the IoD, specifically, on the drone battery efficiency. In summary, *IPCR* is interlaced with the results of *IFDR*. We can note a similar pattern in the bar results, comparing the charts in Figs. 9 and 10. These patterns can be identified because the movement of the drone propellers affects energy consumption more than the transmission and reception of wireless messages [32]. Hence, the higher the *IFDR*, the higher the *IPCR*.

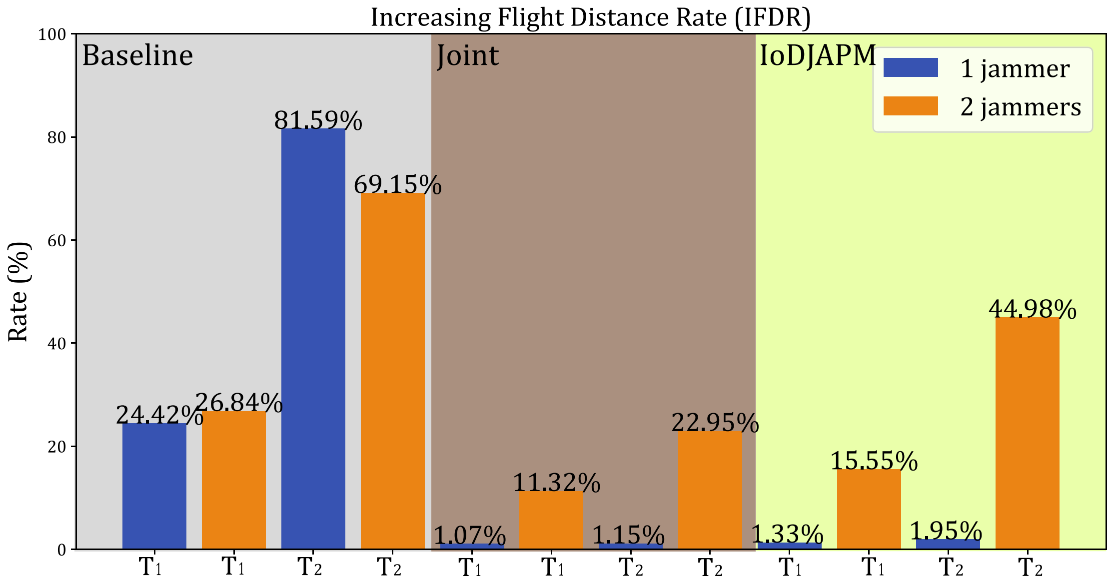


Fig. 9. Results of *IFDR* for each mechanism, topology, and the number of jammers, considering the trips not canceled.

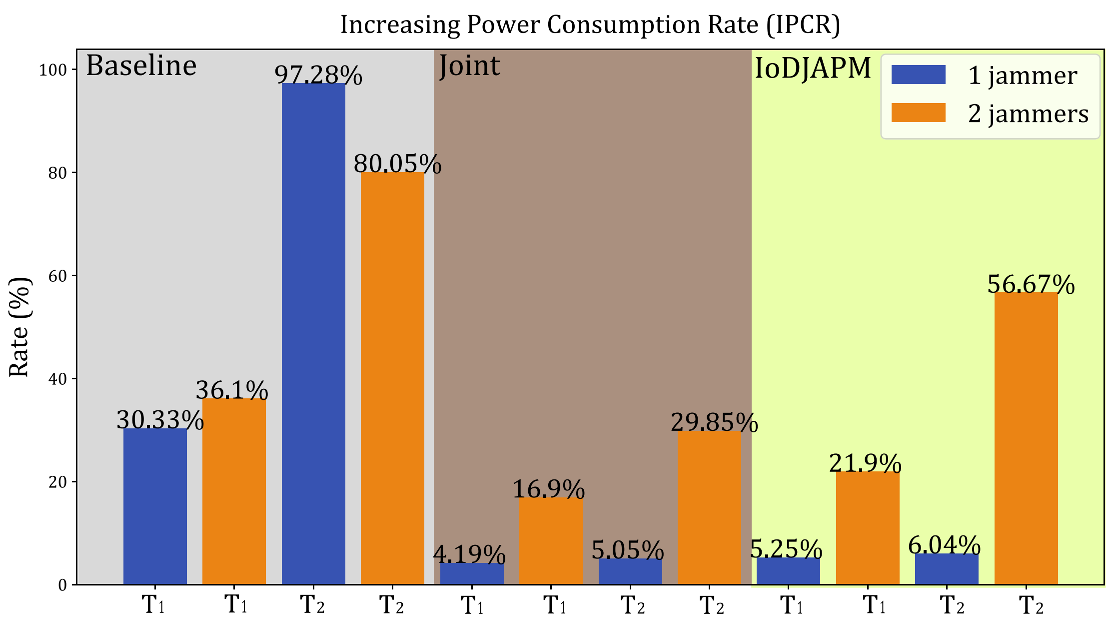


Fig. 10. Results of *IPCR* for each mechanism, topology, and the number of jammers, considering the trips not canceled.

Likewise discussed in the previous results, IoD-JAPM overcomes the baseline mechanism [27] in all scenarios, with discrete increasing power consumption for scenarios with one jammer. Nonetheless, the *IPCR* is higher than 50% in T_2 with two jammers. Although the rate of reformulated path planning is not higher (about 20% on average), the flights redirected to *vertiports* sustain a significant power consumption.

Comparing IoD-JAPM with the Joint approach [18], the results follow the same behavior presented in *IFDR*, which can be explained by the same reasons. Although the power consumption of IoD-JAPM is considerably higher in the more restricted scenario, the *DAPPR* must be taken into account, where IoD-JAPM overcame the Joint Approach in all scenarios.

These aspects reinforce the challenge imposed by restricted topologies. Even with IoD-JAPM, the lack of available pathways directly impacts on the distance that drones fly to complete their itinerary and, therefore, the power consumption. To mitigate these factors, *vertiports* can be placed in strategic regions to cover the airspace optimally. However, this placement involves several aspects since they are commonly deployed in buildings or dedicated areas. All in all, new studies must be conducted to investigate it properly.

6.5. General discussion

The drone traffic control through well-defined airways is undoubtedly fundamental to IoD. With the constantly growing of drone-based services, it is mandatory that a network authority manages and control the drone flight, imposing flyable boundaries. Although the concept of airways promotes several advantages, it represents a risk to avoiding JA, severely affecting the drone performance, as discussed in this study.

In a nutshell, the obtained results pointed out a deep relation between the impact of JA on the drone path planning and the robustness of the airway topology. The more restricted the topology, the more impact JA causes in the IoD: the drone trajectory deviates considerably compared to its original path, increasing the flight distance and power consumption.

IoD-JAPM can mitigate this impact, overcoming our baseline mechanism [27] in all scenarios considering the evaluated metrics. Indeed, IoD-JAPM significantly decreases the number of affected path plannings (mainly those canceled due to the region affected by JA), the increasing flight distance, and the power consumption of the reformulated path plannings.

Regarding the approaches designed primarily for airspace "free-to-fly", the performed drone maneuvers can outperform IoD-JAPM regarding the flight distance and power consumption in scenarios with restricted topology. Nonetheless, these approaches cannot provide a valid trajectory when the jamming signal is strong, as pointed out by our comparative evaluation with the Joint approach [18].

Restricted airway topology against robust jammers represents a profound challenge that must be investigated further. Even with the application of proper anti-jamming mechanisms, an IoD airway topology with few available aerial pathways and few parallel altitudes is affected by strong jammers severely. Depending on the strategic position of these attackers, the protection mechanism (e.g., IoD-JAPM) cannot mitigate the attack correctly, and, therefore, the network can be compromised, affecting the drone flight.

As cars can wait on the roadsides when an accident occurs, drones may have a proper place to stay until the network recovers from the attack. *Vertiports* are proper PoIs to serve as waiting regions. However, they must be placed to optimize access and drone allocation, involving several factors. Hence, the *vertiports* coverage and placement are open challenges to investigate further.

7. Conclusion

In this study, we investigated the impact of JA on drone path planning, and, therefore, the drone trajectory on the IoD. We discussed thoroughly that existing solutions could not adequately meet one of the main IoD characteristics: the drone traffic control through well-defined airways, where the drones fly over constrained airspace.

Hence, we proposed the IoD-JAPM, an airway-aware protection mechanism against JA on the IoD. This mechanism can mitigate the JA impact by searching for an available altitude to communicate with the ZSP without JA interference. To address it, our solution takes advantage of the unused airspace between two or more parallel altitudes over different airways without violating the airways boundaries, avoiding collisions with other drones or obstacles (e.g., buildings). Also, IoD-JAPM redirects the drones to *vertiports* when generating a new path planning is not possible due to JA, keeping the drone safe until the attack is adequately neutralized.

We conducted an experimental evaluation through simulations, comparing IoD-JAPM with our baseline solution [27] and an adapted existent approach [18] that consider the airspace free to fly. We considered environments with different airway topologies and a different number of jammers performing attacks.

IoD-JAPM overcomes the baseline solution in all scenarios, mitigating the effects of JA over the path plannings, causing few reformulations or cancellations. Furthermore, IoD-JAPM causes a slight increase in the drone original flight distance, leading to better power consumption management.

Compared to the "free-to-fly" approach [18], IoD-JAPM presented a similar increase in flight distance and power consumption for robust topology. In constrained scenarios, IoD-JAPM is outperformed in these characteristics, which is an expected behavior since the "free-to-fly" approach does not reformulate path planning. This issue directly impacts the drone trajectory in environments with high jamming signals, where the method cannot process a valid trajectory deviation, leading to several flight cancellations. Thus, our evaluation reinforces that these approaches cannot fit properly with the deployment of IoD environments following well-defined airways.

Also, the results reveal novel challenges in this field. IoD environments with restricted airway topology (i.e., few available pathways and few parallel altitudes to fly) can be severely impacted by JA, even with the application of IoD-JAPM. This impact is related to the strategic position of the attacker. An interesting way to mitigate this impact is to model strategies for the optimal deployment of *vertiports*. In this strategy, the *vertiports* are deployed in points to optimize the environment coverage and the drone flight distance.

In future directions, we plan to explore the opportunities and investigate the current challenges, summarized as follows. (i) Design novel strategies to optimize the deployment of *vertiports* and energy-related constraints, for instance, based on ML approaches [33,34]. (ii) Study the computational, financial, and legal factors to deploy IoD environments with robust airway topology. (iii) Apply the mechanism in a variety of topologies to evaluate the provided protection in a broader scope. (iv) Apply and evaluate IoD-JAPM in real drone-based mobile environments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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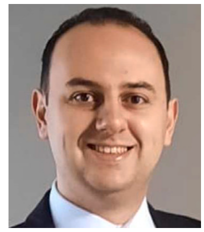

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